

Article

Assessment of genetic diversity in chilli (*Capsicum annuum* L.) genotypes for maturity duration, yield, and quality attributes

Tanveer Ahmad¹, Noor Huma², Muhammad Huzaifa³, Israr Ali⁴, Javeria Azam⁵, Inaba Hawraa⁴, Ghulam Mujtaba ^{6,*}

¹ Pir Mehr Ali Shah Arid Agriculture University, Faculty of Agriculture, Department of Plant Breeding and Genetics, Rawalpindi, Pakistan

² National Agriculture Research Center (NARC), Climate Energy Water Research Institute, Islamabad, Pakistan

³ Scotland's Rural College (SRUC), Craibstone Estate, Ferguson Building, Aberdeen, UK

⁴ Pir Mehr Ali Shah Arid Agriculture University, Faculty of Agriculture, Department of Horticulture, Rawalpindi, Pakistan

⁵ Quaid-i-Azam University, Faculty of Biological Sciences, Department of Plant Sciences, Islamabad, Pakistan

⁶ Pir Mehr Ali Shah Arid Agriculture University, Faculty of Agriculture, Department of Agronomy, Rawalpindi, Pakistan

Abstract

Chilli also known as hot pepper is an essential vegetable that is grown all around the world. It is a good source of minerals and vitamins but its yield in Pakistan is lower than average yield of the world. Chilli has been facing several edaphic factors such as temperature fluctuations, drought and soil conditions in Pakistan. Consequently, its growth and yield are reducing significantly. The impact of climate conditions on chili production is a critical concern, particularly due to the extended growth period of chili crops, which exposes them to a heightened risk of insect infestations and diseases while they remain in the fields. The current study was conducted to evaluate chilli germplasm collection comprising of 50 genotypes for maturity duration and yield. The experiment was conducted in Augmented Block Design (ABD) with two high yielding check varieties. The data regarding various morphological (days to 50 % flowering, days to 50 % fruiting, plant height, etc.) and quality (guaiacole peroxidase, ascorbate peroxidase etc.) traits was collected. The results were subsequently analysed using the ANOVA technique. The LSD method was used for the comparison of means. The correlation among various metrics was also determined using Pearson's correlation. The genetic variation across 50 genotypes of chilli was investigated by doing analysis of variations for each parameter. A highly significant variable with varieties is shown in the analysis of variations among every single parameter of the chilli genotypes when intervention adjustment is evaluated alongside all parameters. The research made it very clear that overall fruit output is critical, thus finding a genotype with a high yield is essential. The genotypes with the greatest recorded yields were R-56 (4381.6), R-49 (1569.1), R-74 (3597.4), and R-83 (1068.3). This finding suggests that four different genotypes have the ability to mature early and produce a lot of fruit.

Article History

Received 23.07.2026

Accepted 17.09.2026

Keywords

Chilli (*Capsicum annuum*); germplasm evaluation; augmented block design; morphological traits; ascorbate peroxidase; genotype selection

* **Corresponding Author** Ghulam Mujtaba  Pir Mehr Ali Shah Arid Agriculture University, Faculty of Agriculture, Department of Agronomy, Rawalpindi, Pakistan

Introduction

Chilli (*Capsicum annuum* L.), Paprika, and pepper are all members of the Solanaceae family of plants, and chilli is a diploid and its chromosomes are 24 in numbers. Chilli was originated South America, this crop is grown in tropical and subtropical regions globally. According to Padilha and Barbieri (2016), five species of chilli peppers *C. annuum*, *C. frutescens*, *C. baccatum*, *C. chinense*, and *C. pubescens* are considered domesticated. Bell pepper cultivation, sweet peppers hot pepper, and chilli are all names for the same vegetable crop (*Capsicum annuum* L.). As a matter of fact, it is an essential part of the diets of a quarter of the global population. The culinary, pharmaceutical, and cosmetics sectors which use it for their purposes. Although it turns woody when fully grown, chilli is most often cultivated as a crop that grows on beds. Over 1.9 million hectares were devoted to chilli farming in 2017, contributing to a global production of more than 36 million tonnes. From an area of over 1.3 million ha, Asia managed to harvest almost 24 million tonnes. Turkey, China, Mexico, Indonesia, and Spain are the world's five leading chilli producers. Half of the world's green chillies come from Asia. According to the Food and Agriculture Organisation of the United Nations (2018), 40% of the world's chilli production and 17% of total exports come from India.

Both *C. annuum* and *C. frutescens* are extensively cultivated in Pakistan. As per Hussain and Abid (2011), the province of Sindh ranks first among Pakistani chilli growers, with Punjab and Balochistan following closely after. In 2019 and 2020, Pakistan produced an average of roughly 141.5 thousand tonnes of chillies per year, making it the leading producer in the world. Nevertheless, it managed to produce 105.4 thousand tonnes between 2020 and 2021. During 2019–20, Pakistan's chilli-producing area was 60752 ha, while in 2020–21, it dropped to 46826 ha (GOP, 2020).

The overall amount of chillies grown in Pakistan, though it is declining from past few years. The worldwide chilli market is not captured by Pakistan (GOP, 2020). Despite being the world's fourth-largest chilli producer, Pakistan's exports are limited because of poor yields and declining quality. Abiotic factors include things like water scarcity, salt, extreme heat or cold, and poor soil quality, but biotic factors include things like pests and diseases. Plants undergo physiological changes in response to each of these circumstances. Chilli peppers have had a high level of genetic variability, which made it easier to select for better genotype where ever the desired characters such as early maturity and high yield potential are required. Yatung et al., (2014) reported that such traits are useful in enhancing effective crop management and productivity for improved income. Several agricultural and external parameters influence the development of chilli plants whereby, maturity time stands out as the most critical factors.

Crop production emphasizes on early maturity and high yields. One of the benefits that early-maturing cultivars afford is that their life cycle can be concluded prior to their being detrimentally affected by climate stress or insects/pests (Mo et al., 2017). Early maturing cultivars on the other hand the researchers, Zhao et al. (2022) suggests that cultivars improve yield through multiple cropping opportunities within a single season. They also have high yields because early-maturing crops do not struggle with weeds and other plants that are (Djaman et al., 2020).

Yields and maturation are critical elements of agricultural production. Early-maturing cultivars has several advantages, including the ability to complete their life cycle prior to adverse weather conditions or significant pest infestations (Mo et al., 2017). Furthermore, they facilitate multiple cropping cycles within a suitable climate, hence enhancing

production. These crops have benefit to other biotic and abiotic factors which causes less yield comparatively. This research explored the best screened genotypes which very helpful to enhance food availability to everyone for a whole year and suitable for production purposes and also for future research breeding programmes for varital development and other trials as well, this current research is very helpful for expanding crop production in the future to satisfy increased food demands using promising early-maturing and high-yielding cultivars.

Material and methodology

This study was initiated after the PMAS Arid Irrigation University Rawalpindi Department of Plant Breeding and Genetics offered fifty genotypes. These genotypes were brought to us by former departmental research students. These genotypes have been maintained by the department from a previous research student. The experimental site was the Horticulture Experiment Area at Rawalpindi's PMAS-Arid Agriculture University. Using seedling trays, the nursery was constructed. Chilli genotype seeds were scattered to a depth of 1 cm in trays that has been previously filled with peat moss. The nursery has been established on a field bed utilising Augmented Block design 45 days after sowing. In a total of four blocks, fifty genotypes were examined, two of which were controls. There are fourteen entries altogether (12 genotypes + 2 check varieties), with the last two varieties appearing twice in each block. The overall length of the row was 5 meters, with 75 cm between the rows and 45 cm separating the plants inside of each row. To highlight the problem of early growth in chillies, researchers looked at a number of quantitative and qualitative variables.

Quantitative characteristics

The plant's primary stem was measured from its base to the axil of the uppermost leaf to ascertain its height. We calculated the mean of the data from five randomly chosen plants. For each genotype, five plants were chosen at random and their stem diameter taken from the middle area to the first bifurcation. The average diameter was then calculated. Starting on the day of transplantation, data was collected until half of the plants showed signs of open flower. Five plants of each strain were chosen at random and their days to blooming were recorded. The average was then calculated. The number of days until 50% of the plants yielded mature fruits was calculated from the date of transplantation. Five plants were chosen at random from each genotypes to determine the number of days, before the average was determined.

A tape with measurements was used to measure the data at its broadest point. For each one of them, five plants were selected at random, and the mean was then calculated. Using a tape to measure with we took the average fruit length from five plants belonging to each genotype that were chosen at random. We counted the amount of ripe fruits from a single plant. For each one of them, five plants were selected at random, and the mean was determined. Five fruits from each genotype were counted for the total quantity of locules after collection, and an average was then calculated. After the fruit of each breed was picked, its seeds were removed. After the hundred beans were weighed using a weighing scale, the data was recorded using a seed counter machine. According to Steel et al. (1997), R-studio was used to conduct analysis of variance (ANOVA). Then, following the procedure outlined by K. Ruben Gabriel (1971) for least significant variation (LSD), correlation was calculated.

Results

ANOVA analysis

Table 4.1 demonstrated the outcomes of the variance analysis for 50 genotypes that were examined in 2023. Fruit width, fruit length, proline content, length of plant, total plant yield, and a hundred seed weight (HSW) showed very significant results up until 50% flowering. Results were statistically significant for the following parameters: fruit yield per plant, the stem’s diameter leaf area index, days to producing fruit, and number of locules.

Table 4.1 Variance analysis of multiple variables across different chilli genotypes in 2023 and 2024. While stem diameter, leaf area index, fruit diameter, and fruit length showed highly significant results for some of the chilli genotypes, other aspects such as days to flowering, days to fruiting, the total number of locules, height of plant, amount during fruits according plant, and total yield of plants showed significant results for all fifty genotypes.

Table 1. Table 4.1: Variance analysis for fifty chilli genotypes for early maturity and yield in the years 2023–24.

S O V	Parameters		Block. unadj	Trt. Adj	Control	Control/ aug	Residual
DF			4	51	1	49	4
Mean Square	D F F	2023	318.34	43.08*	40.00	43.15*	0.000
	D F F	2024	280.667	44.38*	62.500	43.985*	0.000
	D F	2023	234.782	51.652*	90.000	50.885*	0.000
	D F	2024	246.958	48.696*	62.500	48.420*	0.000
	F L	2023	3.6661	6.3746**	0.1000	6.5001**	0.000
	F L	2024	4.3670	6.2848**	0.5063	6.4004**	0.000
	F W	2023	1.8556	2.2551**	0.4000	2.2922**	0.000
	F W	2024	1.67712	2.24395**	0.58403	2.27715**	0.000
	N L	2023	0.38525	0.16075*	0.00000	0.16397*	0.000
	N L	2024	0.282623	0.161008*	0.017361	0.163881*	0.000
	T PY	2023	23747	25697**	355637	19098**	0.000
	T PY	2024	565666	1091344*	6370434	985762*	0.000
	P H	2023	297.100	94.613**	40.000	95.705**	0.00
	P H	2024	279.798	94.628*	30.625	95.908*	0.00
	S D	2023	0.076417	0.022435*	0.025000	0.022383*	0.000
	S D	2024	0.099111	0.023037**	0.008507	0.023327**	0.000
	LAI	2023	0.37012	0.16421*	1.11309	0.14523*	0.00000
	LAI	2024	0.37235	0.16104**	0.85242	0.14721**	0.00000
	N FP	2023	28243.3	5511.7*	490.0	5612.2*	0.0
	N FP	2024	28602.5	5507.2*	562.5	5606.1*	0.0
	H SW	2023	0.033332	0.012739**	0.001000	0.012974**	0.00000
	H SW	2024	0.052042	0.013504*	0.001174	0.013751*	0.00000
	N C	2023	12.5000	4.2075*	10.0000	4.0917*	0.0000
	N C	2024	12.50000	4.2075*	10.0000	4.0917*	0.0000

Factors like plant height and the diameter of the stem, vegetative growth, and overall robustness—the plant's ability to handle a heavy fruit load—must be understood in order to utilise genotype variability and genetic ability in chilli cultivation initiatives towards early maturity varieties with high yields. In order to synchronise output with market demand and climatic conditions, DF and DFRs play a significant role in establishing the maturity period. The commercial yield and customer acceptance are affected by fruit traits including FW, FL, and FW. Additionally, for both the generation of seeds and fresh market usage, NF, NL, and HSW together characterise viability and productivity (Yatagiri et al. 2014-1

One of the most important factors that determines a plant's yield is its height. There was a large range of plant heights across all genotypes. The average plant height in 2023 for all fifty chilli genotypes is shown in Table 4.2. Previous records for plant height include 86.78 cm (CGP-19), 84.28 cm (R21), and 84.28 cm (CGP-18). For R-71 and R-30, the lowest recorded height of the plants were 36.28 cm and 40.78 cm, respectively. The average plant height in 2024 for fifty different chilli genotypes is shown in Table 4.3. The shortest plant height reported was 19.28 cm (R-71) and the tallest was 91.90 cm (CGP-19). The genetic variety in chillies causes a wide range of heights among genotypes. Genetics determine, essentially, how tall a plant will grow. The authors Memon, Ahmad, Depar, Pathan, and Ibrar (2021) provided an explanation for the variation in genotype-environment interactions. When evaluating plant height, it is crucial to include genotype and treatment effects, as shown in these trials. The results highlight the need of understanding the relationship between genetic and environmental impacts on breeding attempts and technical practices, and they also give a better platform for future research into the genetic factors determining plant height in chilli genotypes.

To further understand how crop plants respond to abiotic stresses, the size of the stem is an important measure. Regarding stem diameter, there is substantial heritable variation across all genotypes. Table 4.1 shows the average stemmed diameters of fifty different chilli genotypes in the year 2023. The maximum stem diameters were 1.18 cm for CGP -2 and 0.93 cm for R-21, while the minimum recorded stem diameters were 0.28 cm for CGP -3 and 0.18 cm for R- 61. Table 4.2 displays the average stem size of fifty chilli lines form the year 2024. The maximum diameter was 1.28 cm for CGP-2, while the minimum was 0.34 cm for R-83. A typical plant thickness feature, the lateral the stem, may even explain why different chilli lines have different diameters. Plant thickness is significantly influenced by merismatic activity. Beyond this, the measurement of diameter of stems can be influenced by external conditions including temperature and the availability of water. Below is the mean curve for stem diameter for the 50 chilli genotypes that were studied.

An important metric for the young age examination is the duration to attain 50% days to flowering. Concerning the number of days to 50% flowering, there was a great deal of variation across all genotypes. The average values for the period to 50% flowering in 2023 for 50 different chilli genotypes are shown in Table 4.2. After 60.92 days (CGP-16) and 76.92 days (CGP-10), the shortest time to 50% flowering was recorded at 60.92 days. On the other hand, after 32.42 days, both C-1 and R-67 reached 50% blooming.

As shown in Table 4.3, the average value for the period from to 50% flowering in 2024 over 50 different genotypes of chilli peppers. It took 73 days (CGP-10) for the plant to attain 50% blooming, whereas the shortest duration was 32 days (R-9). It is likely that environmental factors, which speed up early flower blooming as the crop develops, are the primary cause of the difference in days to 50% flowering across chilli genotypes. According to Yatagiri, Sanap, and Telugu (2017), additive gene activity is a good way to estimate yield and also accounts for diversity.

A higher yield is possible when early fruiting is coupled with some crucial qualities, one of which is the number of days to fruiting. The number of days till fruiting showed substantial genetic diversity across all genotypes. Fig. 4.2. Values averaged across fifty different chilli strains for the number of days before harvest in 2023. Minimum days to fruiting were 47.28 (R-67), followed by 47.28 (R-62) while the maximum days to fruiting were 94.78 (CGP-10), 71.78 (CGP-16), and 47.28, respectively.

In terms of the number of days it takes for 50 different varieties of chillies to ripen in 2024, the average value is shown in Table 4.3. The 95(CGP-10) variety accomplished the most fruiting days, while the R-83 variety achieved the fewest, at 50 days. The environmental conditions, like a lack of fertiliser and irrigation, play a major role in the difference in days to blooming across chilli genotypes. These factors help the plants move from the vegetable to the growth phase more smoothly (Datta & Jana et al., 2016). Research indicates that fruit length is a significant metric for assessing overall size and form. Fruit length exhibited significant genetic heterogeneity among all genotypes. Table 4.2 displays the average fruit length measurements for 50 chilli genotypes in 2023. The maximum fruit length observed was 16.47 cm (CGP-20), followed by 13.87 cm (CGP-5). The minimum values were 4.42 cm (R-62) and 3.47 cm (CGP-10), respectively.

Fig 4.2. Mean values for fruit length in 2024 over 50 different chilli genotypes. The greatest width of the fruit was 16.65 cm of CGP-20, and the least width of the fruit was of 3.70 cm of CGP-10. All the genotypes differed significantly in fruit length mean. This is due to the fact that fruit elongation is brought about fruit by mere fact that meristem cells, pollination, and flower quality, among others. Fruit length as well as other yield attributes depends with pollination and the growing environment; but yield characters are genotypically different (Reddy et al., 2017). Perhaps one of the most important aspects of yield is the size of the fruit, which may be defined here by its width. The width of the fruits within all of the genotypes varied widely. The fruit width for fifty different chilli varieties for the year 2023 is also depicted in the following Table 4.2. The broadest fruits had a width of 9.24 cm [CGP-12], the second broadest was 8.94cm[CGP-10], the narrowest 1.60 cm [CGP-7] and 2.30 cm [CGP-4].

Table 4.3 This shows the projected whole-plate mean width of 50 different chilli genotypes in year 2024 The widest chilli fruit was genotype CGP-12 with 8.43cm while the thinnest was CGP-7 with a width of 1.25cm. The fruit width for some degree had some variation to the extent in each genotype. There are proportions influencing the width of the cell, which will result to an increase or decrease of the width of fruits. According to Kumar et al. (2019), other yield attributes in chilli, including fruit width were noted to be responded to genetics variation among the experimental characters. Any other factor that affects the growth and development hormones, and other genetic control the cell elongation affects the diameter of the fruit, noticeably. The research conducted showed that fruit diameter was influenced by characteristics of the surroundings Including water and temperatures (Yadav et al. 2016). These studies elucidate the factors accounting for width of the fruit and, thus, total prospects of producing chillis in light of environmental conditions, hormonal balance, cell growth and genetic differences.

This study showed that the LAI was significantly different between the genotypes in the different stands. Mean values of the index for the leaf area of fifty different chilli genotypes for the year 2023 have been presented in table 4.2. R-21 had only the maximum index of leaf size 2.88 cm², while the minimum was identified in R-32 as 0.14 cm², CGP-12 as 1.64 cm² and R-62 as 0.22 cm². Table 4.3 shows the average 2024 leaf area index for fifty different chilli genotypes. The range of the leaf area index ranged from 0.04 cm³ at the

bottom to 2.61 cm³ at the top (R-111). The advancement and growth of plants nutrients can have a substantial impact on the difference in leaf area index across chilli genotypes. Difference between these genotypes for this parameter nicely described and clarified by environment and the genotypic factors, according to Tiwari et al. (2017), who discovered that various factors, including genetics, environmental conditions, and fertilisers, impact the leaf area index.

Researchers observed a direct correlation between the fruit output of chillies and the quantity of fruits. In terms of fruit yield, there was a great deal of highly significant genetic variation across all genotypes. Table 4.2 shows the average quantity of fruits for 50 different chilli varieties in 2023. In terms of total fruits, the highest recorded number is 254 (R-53), followed by 248 (R-62 check), and the lowest recorded number is 18 (CGP-11), followed by 20 (CGP-14). The mean numbers for fruit weight throughout the year 2023 of 50 chilli genotypes are shown in Table 4.16. The range of documented fruits was from 14 (CGP-11) to 221 (R-53). Climate, soil nutrient deficiencies, and biotic stresses are the main causes of the variation in fruit amount among chilli varieties. These results show resemblance with the Mantja et al. (2011) finding.

Chilli fruits which were produced have proportional with number of locules. Regarding the number of locules, there was a great deal of genetic variation across all genotypes. Table 4.3: Values averaged for 50 chilli genotypes in terms of locule count in 2023. Among the recorded locule counts, 4.07 (CGP-10), 3.27 (CGP-12), and 1.07 (CGP-2) were the highest, while 1.07 (CGP-7) was the lowest. Using 50 genotypes as an example, Table 4.18 shows the average value for locule frequency in 2024. The highest measured variety of locules was 3.73 (CGP-10), while the lowest was 1.55 (CGP-5). Chilli genotypes differ in the quantity of locules, which could be explained by the lateral meristem. Despite establishing a strong correlation between locule quantity and fruit weight, Reddy et al. (2017) found a highly valuable and practical genetic diversity of locule count in 30 different chilli genotypes. In a study involving fifty different chilli genotypes, Kumar et al. (2019) discovered a direct correlation between the quantity of locules and fruit yield ($r = 0.87$).

One way to tell which cultivars produce more than average is to look at their overall plant yield. All genotypes showed substantial genetic variability in total plant output. In Table 4.2, we can see an average total fruit yield of fifty different chilli genotypes in 2023. With a total of 729.85 g (R-81), the highest reported yields of fruits was followed by 685.83 g (R-53), the second highest. Total fruit yields as low as 116.03 g (R-62) and as high as 125.45 g (R-36) were recorded. In Table 4.3 Yield averages for 50 different chilli genotypes in 2024. Out of all the plants, the one with the highest yield was 438.06 (R-86), although the one with the lowest yield was 492.2g (R-19). When all the elements happening in the fields, such as climate change and a decline in overall fertility, are considered, the disparity between total yield of crops and total fruit yield remains. This is the explanation for it. The study's depiction of substantial genetic variation is in line with previous results from other researchers who have suggested that the current genetic variability could be used to generate chilli cultivars with higher yields. Chilli production might be made more economically viable and food secure through selective breeding of genotypes with favourable yield properties, which would lead to increased productivity overall. With global demand for chilli peppers on the rise and climate change proving to be a real challenge, this is more important than ever. Golparvar, Mottaghi, and Lotfifar (2011) also found comparable results.

Chilli fruit has very tiny seeds which is light in weight and hundred seed weight, there was a highly substantial genetic diversity across all genotypes. The average values of fifty different chilli genotypes for one hundred seed weights in 2023 are shown in Table 4.2. In CGP-13 and CGP-5, the highest hundred weight of seeds was 0.74 g, whereas in R-73 and R-9, the lowest hundred seedlings weight was 0.20 g. Data from 50 different chilli genotypes, averaged over 100 seed weights in 2024 (Table 4.3). A hundred seed weight as low as 0.28 g (R-32) and as high as 0.90 g (CGP-13) were recorded. Although there are limitations to this relationship, especially when it comes to germination, there is a positive association between seed weight and seed size.

In 2023, the amount of huge seeds used per flowering area would be increased, which means that one hundred seeds would weigh more, as per Azmik, Sembiring, and Manik. The data show that different genotypes of chilli have different seed weights. In most cases, higher HSW values indicate better seed quality and more room to grow. According to Singh et al. (2009), genotypes with higher HSW values can be selected to enhance seed performance and total productivity.

There was an obvious variation among glutathione peroxidase types. Table 4.2 displays the average values of glutathione peroxidase across fifty different chilli strains. At their peak, glutathione per levels were 46.92 (R-62), and they were second-highest at 45.92 (R-74). At their lowest, they were 18.42 (R-76; CGP-11). Though R-76 and CGP-11 exhibited the lowest GPX levels (18.42 and 18.92, respectively), they might be more susceptible to oxidative damage, which could change development and yield in stressed environments. These findings corroborated previous studies that linked higher GPX activity to improved stress tolerance in plants and laid the groundwork for selecting genotypes with GPX-based growth profiles. In the long run, this variation in gene expression could be utilised to enhance the nutritional content and stress resistance of chillies, which would help with production resilience and long-term vi in light of climate change. Similar results were found by Ramalho, and Lidon (2010).

Ascorbate peroxidase is a crucial indicator for identifying cultivars with high output potential. The genotypes of ascorbate peroxidase exhibited significant genetic diversity. Table 4.2 illustrates the mean values of fifty distinct chilli genotypes for the specific form of as under investigation. The lowest ascorbate levels were 10.57 (CGP-9) and 11.57 (R-26), while the highest were 54.07 (R-62) and 54.07 (R-73). The peroxide can be reduced to water using ascorbate, a reducing agent. As a result of dominant gene expression, ascorbate peroxidase levels in chillies exhibit significant variation among genotypes. Numerous studies have established a correlation between elevated APX activity and stress tolerance in several plant species; thus, selecting genotypes with high APX levels may serve as an effective way to significantly improve crop resilience and productivity.

This feature can be further utilised in breeding efforts to create strains of chilli that are resistant to stress and produce more fruit because the primary gene for activity has a substantial effect. Chilli peppers with high APX activity genotypes have better chances of survival in harsh environments because they increase yield and nutritional value. This, in consequence, is in line with the rising tide of climate-resilient agriculture, which promises both food security and sustainability issues at the same time. The study was conducted by Kumar et al. in 2011.

Table 2. Means of attributes of 50 chilli genotypes during year 2023

Varieties	PH	SD	DFF	DF	FL	FW	LAI	NFP	NL	TPY	HSW	GPX	APX
CGP-10	54	0.7	75.0	93.0	5.3	8.6	0.8	33	3.9	201.7	0.58	33	41
CGP-16	63	0.4	59.0	70.0	5.8	3.7	0.5	99	1.9	250.8	0.48	34	39
CGP-15	53	0.6	56.0	69.0	6.6	5.3	0.8	73	1.9	280.7	0.30	27	35
CGP-12	72	0.6	55.0	69.0	5.6	8.9	1.5	37	3.1	204.0	0.61	34	23
CGP-7	63	0.7	55.0	68.0	5.8	1.5	0.3	77	1.8	153.5	0.54	31	36
CGP-3	68	0.3	53.1	68.0	11.1	4.6	0.4	43	2.0	228.5	0.58	41	44
CGP-13	79	0.7	53.0	63.0	4.8	6.4	0.6	74	3.0	188.4	0.73	36	42
CGP-19	87	0.5	53.0	69.0	9.1	2.7	1.2	51	2.1	160.5	0.51	27	33
CGP-11	57	0.6	52.0	65.0	6.3	6.1	0.3	17	2.9	175.5	0.56	21	25
CGP-20	78	0.7	52.0	67.0	16.4	2.8	0.5	31	1.9	198.9	0.60	32	37
CGP-4	55	0.4	50.3	64.0	7.3	2.2	1.0	66	2.1	162.8	0.52	36	42
CGP-14	64	0.5	50.0	67.0	6.6	6.9	0.7	19	3.1	147.1	0.58	39	44
CGP-9	64	0.6	45.0	56.0	6.5	8.4	1.1	57	2.9	247.4	0.56	27	12
R-73	59	0.5	43.0	55.0	8.3	3.0	0.5	231	2.0	293.7	0.24	44	54
R-44	63	0.4	42.0	59.0	8.1	4.0	0.7	167	2.0	273.0	0.49	32	37
CGP-2	75	1.2	41.6	54.0	7.5	4.7	0.6	41	1.8	286.1	0.57	38	42
R-36	47	0.3	41.0	58.0	6.3	4.5	0.3	69	1.9	122.5	0.29	25	31
CGP-18	82	0.8	40.0	53.0	10.4	6.2	1.2	49	1.9	201.7	0.56	37	44
CGP-5	67	0.4	40.0	55.4	13.9	4.4	0.8	42	2.1	455.8	0.68	23	30
CGP-8	65	0.5	40.0	55.0	9.1	6.2	0.5	52	1.9	244.8	0.61	21	17
R-10	67	0.7	40.0	62.0	5.3	4.9	1.1	153	2.1	272.4	0.50	29	35
R-30	58	0.6	40.0	60.0	4.2	5.2	0.4	125	2.1	440.2	0.30	35	41
R-38	65	0.4	40.0	55.0	3.9	3.8	0.7	263	2.1	578.5	0.32	35	42
R-55	63	0.4	40.0	55.0	8.2	3.2	0.8	163	2.1	372.3	0.53	39	43
R-57	60	0.4	40.0	55.0	9.7	4.2	0.8	225	2.0	244.6	0.59	30	37
R-83	48	0.3	40.0	55.0	12.4	4.0	0.8	167	2.9	239.2	0.40	23	21
CGP-17	65	0.6	39.0	54.0	7.7	5.4	1.0	47	2.9	168.7	0.58	24	28
R-108	49	0.3	39.0	53.0	8.7	3.1	0.3	265	2.1	280.4	0.46	34	39
R-12	55	0.7	39.0	65.0	4.9	3.5	0.9	125	2.1	203.5	0.54	32	37
R-27	46	0.5	39.0	59.6	5.3	4.7	0.8	101	2.0	215.8	0.31	26	12
R-5	52	0.6	39.0	60.0	9.9	5.2	0.6	115	2.1	207.8	0.48	37	44
R-53	57	0.4	39.0	57.0	9.6	4.3	0.4	265	1.8	677.2	0.48	36	42
R-59	57	0.5	39.0	51.0	10.2	3.6	0.3	133	1.8	264.4	0.47	24	27
R-67	53	0.6	39.0	53.0	9.0	3.1	0.1	89	2.0	153.5	0.34	37	48
R-71	40	0.4	39.0	54.0	8.7	3.6	0.9	67	1.9	296.5	0.40	25	30
CGP-1	78	0.7	38.5	53.0	5.4	2.5	0.5	82	1.9	193.6	0.49	37	42
CGP-6	63	0.7	37.0	57.0	8.6	3.4	0.6	107	1.9	332.1	0.37	36	44
R-76	56	0.4	37.0	49.0	7.9	3.7	0.4	215	1.9	356.4	0.49	19	20
R-86	46	0.3	37.0	49.0	5.7	3.5	0.4	87	1.9	274.4	0.37	28	32
R-19	57	0.4	36.0	60.0	5.4	4.2	0.2	57	2.0	164.0	0.50	26	30
R-26	58	0.4	36.0	60.5	6.7	5.2	0.3	80	1.8	366.5	0.26	23	17
R-61	52	0.3	36.0	50.0	4.7	4.9	0.7	198	1.9	243.0	0.65	23	19
R-81	83	0.5	36.0	50.0	12.4	5.0	0.8	155	1.9	239.2	0.50	43	54
R-22	54	0.4	35.0	60.0	5.3	4.2	0.9	134	1.9	288.5	0.55	32	36
R-32	56	0.6	35.0	57.0	6.9	4.2	0.2	97	1.9	185.4	0.24	20	25
R-52	50	0.5	35.0	51.0	8.2	3.0	0.2	155	1.9	161.6	0.39	21	17
R-62	67	0.5	35.0	49.0	3.9	6.1	0.4	263	2.1	109.0	0.45	44	54
R-74	63	0.5	35.0	49.0	10.9	5.3	0.6	133	2.1	603.4	0.60	41	48
R-9	55	0.6	35.0	50.0	7.2	4.7	0.2	215	1.9	260.2	0.20	43	48
C-1	22	0.5	36.0	49.0	7.9	3.7	0.3	18	2.1	222.2	0.41	18	12
C-2	59	0.6	40.0	55.0	7.7	4.1	0.5	21	2.1	599.4	0.43	21	19

Table 3. Means of attributes of 50 chilli genotypes during year 2024

Varieties	PH	SD	DFF	DF	FL	FW	LAI	NFP	NL	TPY	HSW	GPX	APX
CGP-10	53	0.7	73.0	95	3.70	8.4	0.8	33	3.7	121.0	0.59	33	41
CGP-16	61	0.4	56.0	71	6.55	3.4	0.5	99	1.6	225.5	0.47	34	39
CGP-15	52	0.6	54.0	71	7.10	5.1	0.8	73	1.7	168.2	0.31	27	35
CGP-12	71	0.6	53.0	69	6.10	8.7	1.5	37	2.9	122.2	0.60	34	23
CGP-7	61	0.7	52.0	70	6.55	1.2	0.3	77	1.5	138.8	0.53	31	36
CGP3	66	0.3	50.1	69	11.85	4.3	0.4	43	1.7	205.1	0.59	41	44
CGP-13	77	0.7	49.0	67	5.80	6.0	0.6	74	2.6	226.2	0.72	36	42
CGP-19	86	0.5	51.0	71	9.60	2.5	1.2	51	1.9	963.4	0.52	27	33
CGP-11	55	0.6	49.0	68	7.05	5.8	0.3	17	2.6	159.4	0.55	21	25
CGP-20	77	0.7	51.0	68	16.65	2.7	0.5	31	1.9	596.8	0.61	32	37
CGP-4	54	0.4	48.3	66	7.80	2.0	1.0	66	2.6	977.0	0.56	36	42
CGP14	62	0.5	47.0	70	7.35	6.6	0.7	19	1.8	132.5	0.53	39	44
CGP-9	62	0.6	41.0	60	7.50	8.0	1.1	57	1.9	296.6	0.52	27	12
R-73	58	0.5	41.0	57	8.80	2.8	0.5	231	2.8	176.4	0.27	44	54
R-44	62	0.4	41.0	60	8.35	3.9	0.7	167	2.5	819.2	0.44	32	37
CGP-2	74	1.2	39.6	56	8.00	4.5	0.6	41	1.8	171.8	0.53	38	42
R-36	45	0.3	38.0	38	7.05	4.2	0.3	69	1.9	110.5	0.25	25	31
CGP-18	80	0.8	37.0	56	11.15	2.3	1.2	49	1.6	181.9	0.53	37	44
CGP-5	65	0.4	37.0	58	14.65	4.1	0.8	42	1.6	410.5	0.62	23	30
CGP-8	64	0.5	38.0	57	9.60	6.0	0.5	52	1.6	146.0	0.65	21	17
R-10	66	0.7	39.0	63	5.55	4.8	1.1	153	1.8	817.3	0.52	29	35
R-30	57	0.6	39.0	61	4.45	5.1	0.4	125	1.8	132.8	0.33	35	41
R-38	64	0.4	38.0	57	4.40	3.6	0.7	263	2.0	347.2	0.32	35	42
R-55	62	0.4	38.0	57	8.70	3.0	0.8	163	2.0	223.2	0.53	39	43
R-57	62	0.4	39.0	59	9.95	4.1	0.8	225	1.9	733.8	0.57	30	37
R-83	64	0.3	39.0	56	5.65	3.9	0.8	167	1.9	717.6	0.43	23	21
CGP-17	59	0.6	37.0	56	8.20	5.2	1.0	47	1.9	101.6	0.56	24	28
R-108	58	0.3	37.0	55	6.20	2.9	0.3	265	3.9	168.6	0.43	34	39
R-12	57	0.7	37.0	67	5.40	3.3	0.9	125	2.7	122.4	0.52	32	37
R-27	56	0.5	37.0	62	5.80	4.5	0.8	101	1.9	129.0	0.34	26	12
R-5	56	0.6	38.0	61	10.15	5.1	0.6	115	1.9	623.6	0.47	37	44
R-53	56	0.4	38.0	58	9.85	4.2	0.4	265	1.8	203.8	0.48	36	42
R-59	55	0.5	38.0	52	10.45	3.5	0.3	133	2.0	793.4	0.43	24	27
R-67	55	0.6	36.0	56	9.75	2.8	0.1	89	1.7	138.5	0.31	37	48
R-71	55	0.4	36.0	57	9.45	3.3	0.9	67	1.7	266.5	0.44	25	30
CGP-1	54	0.7	36.0	55	5.90	2.3	0.5	82	1.7	116.0	0.42	37	42
CGP-6	53	0.7	35.0	59	9.10	3.2	0.6	107	1.6	199.0	0.36	36	44
R-76	53	0.4	36.0	50	8.15	3.6	0.4	215	1.7	106.3	0.44	19	20
R-86	52	0.3	35.0	51	6.20	3.3	0.4	87	1.7	164.6	0.36	28	32
R-19	51	0.4	35.0	61	5.65	4.1	0.2	57	1.8	492.2	0.53	26	30
R-26	51	0.4	34.0	63	7.20	5.0	0.3	80	1.7	219.2	0.26	23	17
R-61	51	0.3	34.0	52	5.20	4.7	0.7	198	1.9	145.4	0.62	23	19
R-81	49	0.5	34.0	52	12.90	4.8	0.8	155	1.6	438.6	0.53	43	54
R-22	49	0.4	33.0	62	9.20	4.0	0.9	134	1.7	173.2	0.55	32	36
R-32	49	0.6	34.0	58	7.15	4.7	0.2	97	1.7	556.4	0.22	20	25
R-52	48	0.5	33.0	53	8.70	4.8	0.2	155	1.7	969.6	0.32	21	17
R-62	48	0.5	33.0	51	4.40	4.0	0.4	263	1.8	654.4	0.41	44	54
R-74	47	0.5	32.0	51	11.40	4.1	0.6	133	2.8	362.6	0.64	41	48
R-9	44	0.6	32.0	53	7.95	2.8	0.2	215	1.9	234.8	0.23	43	48
C-1	42	0.5	33.0	52	8.65	4.4	0.3	18	3.4	200.1	0.42	18	12
C-2	38	0.6	38.0	57	8.20	5.1	0.5	21	1.9	359.4	0.41	21	19

Discussion

Analysis of correlations

This study shows that the correlation tells us how two separate variables are related to one another. The value of correlation analysis cannot be overstated. It allows us to see how different attributes are related to one another and how strongly these associations interact with one another. According to Kwon and Torrie (1964), this can be deduced. Table 4.1 (a) A study examining the relationship between 50 chilli genotypes and yield in 2023 using Pearson's correlation. Factors such as height of plants, stem diameter, proline content, ascorbate peroxidase, and guaiacol peroxidase catalase, as well as days to 50% blooming, have all been found to be positively connected with fruiting. If the number of days to 50% flowering increases, then it stands to reason that the number of additional features related to this variable will likewise increase, as there is a positive correlation between the two. Time to 50% blooming was a direct function of the parameters, which had a beneficial effect on maturity duration.

There is a substantial amount of research that suggests a positive correlation between early maturity and other phenological features, such as period till flowering. It is likely that a decrease in maturation time is correlated with an increase in flowering time, as Singh (2023) finds a considerable phenotypic relationship of 0.431 between the length of time to reach the advanced green stage and the persistence to 50% blooming. In contrast to the senescence date, the genotypes selected for the flowering date would show a clear maturity date. The farmer, who strives to maximise the agricultural cycle, would also reap the benefits of this.

Days to 50% flowering, fruit length, and average plant production were found to be negatively correlated with one another. The decrease in these variabilities as the numbers of days to fifty % flowering enlarge had an adverse effect on the ripening of the chiles, according to that it has negative correlation. Between the number of locules and fruit weight, there was a positive and large relationship between the duration till fruiting; between fruit length and complete fruit output, there was also a significant relationship. Deshmukh and Chinnusamy (2012) also found things that were similar.

A negative correlation was found between fruit length and both fruit width and locule count, suggesting that as fruit length increased, breadth along with locule count decreased. There was a positive correlation between proline content, fruit breadth, number of locules, stem diameter, leaf area index, and stem diameter. Ascorbate peroxidase levels were significantly inversely related to fruit width.

According to research by Bekele et al. (2022), fruit length has a beneficial effect on fruit yield per plant. This is considered important for evaluating hot pepper yields (*Capsicum annuum* L.). This study's results are in line with those of a chilli research that found a strong relationship between the fruit length and production per plant, indicating that longer fruits could increase yields generally. According to a morphological analysis of pepper varieties grown in Bulgaria, fruit length is a key characteristic that determines how ripe the fruit is and how good it is in general (Tsonev et al., 2017). Despite being a physical feature, fruit length greatly affected the plant's life cycle and maturity date, according to these research.

Stem diameter, proline content, and chilli pepper maturity were positively correlated with locule number. However, there was a strong negative association between the total plant yield and the quantity of locules. Such results have been proven by Amiri,

Bahraminejad, and Jalali-Honarmand. A favourable and statistically significant association is seen between the total plant yield and ascorbate peroxidase, a substance called peroxidase, and catalase. There is a strong positive relationship between plant height and ascorbate peroxidase, guaiacol peroxidase, catalase, stem diameter, leaf area index, and stem diameter.

The leaf area index and peroxidase levels exhibited a favorable correlation with stem diameter. Leaf area index and proline content were not associated with any of the other properties since no correlation was detected between them. While all of the other relationships were negative, ascorbate peroxidase was positively associated with guaiacol peroxidase and catalase.

Figure 4.1 (b) Examination of 50 genotypes in 2024 using Pearson’s correlation Plant altitude stem diameter, number of locules, duration until fruiting, quantity of locules, and hundred seed weight are positively and substantially correlated with the duration until 50% flowering. An increase in the length to 50% flowering improves the linked attributes of that specific trait, according to a positive connection. There is a strong negative correlation between this and both fruit length and fruit yield per plant. A negative correlation indicates that these characteristics decrease with increasing Days to 50% blooming. Here, fruit weight, locule are counted, and seed diameter showed a strong positive correlation with the duration till fruiting. There was a strong negative correlation between the number of cherries per plant and the length till fruiting. Almeselmani, Deshmukh, and Chinnusamy’s (2012) results are similar.

A positive link with total production and plant height was seen for fruit length, and this correlation reached statistical significance. There is a positive and statistically significant relationship between fruit breadth, stem diameter, number of locules, and leaf area rating. But there’s a negative and significant correlation with fruit yield per plant. There was a negative relationship between total fruit output and the number regarding fruits per plant, but a strong positive link between stem diameter and thousand seed weight and the number of locules. In their 2013 study, Amiri et al. confirmed this finding. This study found that the variables stem diameter, hundred seed weight, leaf area index, and plant height were positively correlated. There was a robust positive relationship between stem diameter, leaf area index, and hundred the weight of the seed The number of apples per plant was negatively correlated with it, though. Our results corroborate Keyvan’s (2010) observation of a strong negative correlation between catalase concentrations and fruit yield per plant. Using (a) 2023 and (b) 2024 as examples.

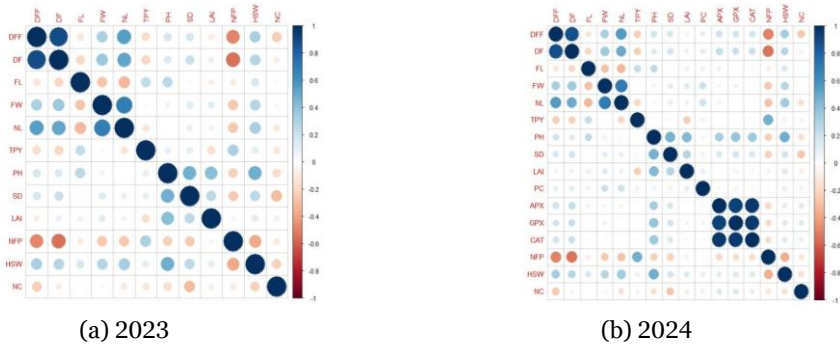


Figure 1. Pearson Correlation Heatmaps Among Morphological, Yield, and Quality Traits of 50 Chilli Genotypes in (a) 2023 and (b) 2024

Figure 4.1 shows the results of a Pearson's correlation analysis conducted on fifty different chilli genotypes. Days to Fruiting (DF): Days to 50% flowering TFF stands for Total Fruit Yield, FL for Fruit Length, FW for Fruit Width, NL for Number of Locates. Plant height (PH): PC stands for proline content, LAI for leaf area index, and SD for stem diameter. It stands for ascorbate peroxidase. GPX stands for guaiacol peroxidase CAT, which is involved with catalase. NFP stands for nodal anthocyanin. HSW stands for hundred seed weight.

Conclusion and implications

The research made it very clear that overall fruit output is critical, thus finding a genotype with a high yield is essential. The genotypes with the greatest recorded yields were R-56 (4381.6), R-49 (1569.1), R-74 (3597.4), and R-83 (1068.3). This finding suggests that four different genotypes have the ability to mature early and produce a lot of fruit.

Competing interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Orcid ID

Ghulam Mujtaba  <https://orcid.org/0009-0009-0968-6874>

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